

Exploring the Possibility of Blood Group Prediction from Fingerprint Images Using Deep Learning

*A Graduation Project Submitted in Partial Fulfillment of the Requirements for
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Intelligence*

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Abstract

Artificial intelligence and deep learning have demonstrated remarkable capability in extracting complex patterns from visual data, with growing applications in the medical domain including diagnostic support and disease classification. One area that remains largely unexplored is the use of biometric data, specifically fingerprint images, as a non-invasive input for predicting biological characteristics such as blood groups.

Blood group identification is critical in medical procedures including transfusions and emergency care; however, conventional methods rely on invasive laboratory tests that limit accessibility in resource-constrained environments. Some studies suggest a potential relationship between fingerprint ridge patterns and blood groups, though this has not been conclusively established.

This study proposes Print2Type, a CNN-based system that explores the possibility of predicting blood groups using fingerprint patterns. A dataset of 6,000 fingerprint images across eight blood group categories was obtained from Kaggle, split 70/30 for training and testing using stratified splitting, with preprocessing including resizing to 224×224 pixels and normalization. Print2Type achieves an accuracy of 89.56%, macro precision of 89.82%, macro recall of 90.13%, and mean AUC of 0.9355, with F1-scores ranging from 0.87 to 0.90. A Vision Transformer model was implemented for comparison, achieving 78.08% accuracy, confirming Print2Type's superior generalization. Results should not be considered a substitute for professional medical diagnosis, as the biological relationship between fingerprint patterns and blood groups remains inconclusive.

Keywords: Blood Group Prediction, Fingerprint Images, Deep Learning, Convolutional Neural Network, Vision Transformer, Print2Type, Biometric Classification, Non-Invasive Medical Prediction

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CHAPTER 1 : INTRODUCTION

Background

The rapid advancement in artificial intelligence technologies has redefined the way data is processed and analyzed. Systems are no longer limited to basic computation but now encompass understanding, pattern recognition, and extraction of complex features from raw data. Among the most prominent areas of this progress is deep learning in image analysis, which has demonstrated high efficiency in extracting hidden features from visual inputs.

In the medical field, artificial intelligence has been increasingly integrated into diagnostic and clinical support systems. Applications range from medical image classification and disease detection to patient monitoring and drug discovery. These developments reflect a growing recognition that AI-based tools have the potential to enhance the speed, accuracy, and accessibility of healthcare services.

Blood group identification is one of the fundamental requirements in many medical procedures, including blood transfusions, organ transplants, and emergency care. Accurate and timely knowledge of a patient's blood group can be critical in life-threatening situations. However, current identification methods rely on laboratory tests that require trained personnel, specialized equipment, and direct physical intervention, making them less accessible in certain environments.

Fingerprints are among the most stable and distinctive biometric features available. They are formed during the fetal stage and remain unchanged throughout a person's lifetime. Some studies in the field of dermatoglyphics suggest a potential relationship between fingerprint ridge patterns and certain genetic characteristics, including blood groups. However, this relationship has not been conclusively established, and its practical exploitation through artificial intelligence remains a largely unexplored direction.

Motivation

The motivation behind this project stems from observing a clear gap between the rapid advancement of artificial intelligence technologies and the slow adoption of these technologies in some basic medical procedures. While it has become possible to analyze complex images and recognize fine patterns

automatically, blood group identification still depends on laboratory tests that do not benefit from this progress.

The project also aims to move beyond a theoretical model by building a complete practical system that combines preprocessing, deep learning, and an interactive user interface. This reflects the importance of transforming AI research into applicable solutions rather than experimental prototypes only.

Accordingly, the motivation is not limited to solving a specific problem, but extends to exploring a new direction that integrates biometric data and artificial intelligence in a practical context.

Problem Statement

Despite the critical importance of blood group identification in the medical field, current methods rely mainly on laboratory tests that require time, specialized equipment, and direct physical intervention. This reliance limits the speed of response in critical situations and reduces the accessibility of these services in some environments.

On the other hand, biometric data such as fingerprint images remain largely unexploited in this field, although some studies suggest the existence of a potential relationship between fingerprint patterns and genetic characteristics including blood groups.

Based on this, the research problem lies in the absence of an intelligent system capable of utilizing fingerprint data to predict blood groups in a fast and non-invasive way. This project aims to address this gap by build Print2Type and transforming it into a practical deployable system.

Project Objectives

Considering the challenges associated with traditional blood group identification methods, this project seeks to explore an alternative approach based on biometric data and deep learning. The project focuses on building Print2Type, an intelligent system capable of providing accurate predictions in a fast and non-invasive manner.

This project aims to:

- Explore the feasibility of predicting blood groups from fingerprint images.
- Build Print2Type, a deep learning system that explores the possibility of predicting blood groups from fingerprint images across eight categories: A+, A−, B+, B−, AB+, AB−, O+, and O−.
- Build a complete pipeline that includes data preprocessing, model training, and evaluation.
- Evaluate Print2Type using standard performance metrics including accuracy, precision, recall, and F1-score.

Contribution

This project provides a contribution at the intersection of biometric data and deep learning by exploring a non-traditional use of fingerprint images for predicting a medical characteristic.

The main contributions include:

- Proposing the use of fingerprint images as a non-invasive input for blood group prediction within a practical framework.
- Build Print2Type, a system capable of automatically extracting visual patterns from fingerprint images and classifying them into eight blood group categories.
- Building an end-to-end pipeline for Print2Type that covers data preprocessing, training, evaluation, and deployment.
- Providing a Gradio-based web interface for Print2Type, deployed on Hugging Face Spaces and accessible publicly without any local installation.

Scope of the Study

This project focuses on classifying fingerprint images into eight blood group categories using Print2Type. The system is trained on a publicly available dataset and does not include fingerprint segmentation, biometric identity verification, or integration with hospital databases.

Print2Type is intended as a proof-of-concept decision-support tool and not as a replacement for professional medical diagnosis. The implementation covers dataset acquisition, preprocessing, model training and evaluation, and deployment through a web-based interface for demonstration purposes.

Report Organization

This report is organized as follows:

- Chapter 2 reviews related literature on fingerprint patterns, fingerprint-based medical prediction, and deep learning techniques in image classification.
- Chapter 3 describes the proposed methodology and system design of Print2Type, including the dataset, preprocessing pipeline, model architecture, evaluation strategy, and web interface.
- Chapter 4 presents the experimental results and discussion of Print2Type performance.
- Chapter 5 concludes the study and provides recommendations for future improvements to Print2Type.

CHAPTER 2 : LITERATURE REVIEW

Deep Learning and Artificial Intelligence

Deep learning is a branch of artificial intelligence that enables systems to learn complex patterns directly from raw data without manual feature engineering. LeCun, Bengio, and Hinton (2015) demonstrated that deep learning models build multi-level representations of data, making them highly effective across a wide range of classification and recognition tasks [1].

Convolutional Neural Networks

Convolutional Neural Networks are the dominant architecture for image analysis tasks. They automatically extract spatial features such as edges, textures, and shapes through convolutional layers, eliminating the need for manual feature extraction. IRJMETs (2024) confirmed that CNN-based models significantly outperform traditional image processing methods, particularly when input data contains fine visual patterns [2].

Deep Learning in Medical Image

The success of CNNs in computer vision has naturally extended into the medical domain. Deep learning models have been applied to a wide range of medical imaging tasks including disease detection, tumor classification, and diagnostic support.

These applications demonstrate that AI-based systems can match or exceed human performance in certain medical image analysis tasks, while also offering advantages in speed, consistency, and scalability. The growing availability of medical image datasets has further accelerated progress in this area, making it increasingly feasible to develop reliable automated diagnostic tools.

Biometric Applications in Medicine

Biometric features such as fingerprints, iris patterns, and facial geometry have traditionally been used for identity verification. However, their potential application in medical prediction has attracted increasing research interest.

Vaidya et al. (2025) demonstrated the ability to use fingerprint structural features such as ridge direction and density as a non-invasive input for predicting biological characteristics including blood groups [3]. A study titled Blood Group Detection Using Fingerprints and Image Processing (2025) further

confirmed that fingerprint-based prediction systems offer a safe and low-cost alternative to conventional laboratory tests, particularly in emergency situations or resource-limited environments [4].

These studies highlight the broader potential of biometric data beyond identification, and motivate the development of AI-based systems that can extract medically relevant information from such data.

Fingerprints and Blood Groups

The scientific study of fingerprint ridge patterns, known as dermatoglyphics, has revealed potential associations between these patterns and certain genetic characteristics. Fingerprint patterns are formed during the fetal stage and remain unchanged throughout a person's lifetime, making them a stable biological indicator.

Susmiarsih et al. (2016) indicated that a relationship between fingerprint patterns and blood groups may exist, though it was not consistent across all studied samples [5]. Manikandan et al. (2019) compared the distribution of the three main pattern types — arches, whorls, and loops — across different blood groups, and found noticeable variation that supports a potential genetic basis for this relationship [6].

In contrast, a study published in Healthcare Bulletin (2022) found no strong or statistically significant relationship between fingerprint patterns and blood groups, suggesting that results may vary depending on sample size and methodology [7].

Ghaffar et al. (2024) conducted a more detailed statistical analysis among students at Majmaah University and reported significant differences in pattern distribution across blood groups. The loop pattern was the most common, followed by whorls and arches, and blood group O was the most prevalent in the studied sample [8].

While the biological relationship has not been conclusively established, the evidence suggests that fingerprint patterns carry potential biological indicators that can be further explored through deep learning techniques capable of detecting subtle visual differences that traditional statistical methods may miss.

Comparison of Related Studies

The reviewed studies collectively highlight the progression from theoretical exploration of the fingerprint-blood group relationship toward practical AI-based prediction systems. While earlier studies relied on statistical analysis of manually extracted features, more recent work has demonstrated the effectiveness of deep learning in extracting complex visual patterns from fingerprint images. However, most existing systems lack practical deployment, and none provide a fully integrated web-based prediction tool. Print2Type addresses these limitations by combining CNN-based classification with a deployed interactive interface.

Table 2-1 Comparison of Related Studies

Study	Year	Focus	Method	Key Finding	Limitation
LeCun et al. [1]	2015	Deep learning	CNN	Multi-level feature learning	General, not domain-specific
IRJMETs [2]	2024	Fingerprint classification	CNN	Outperforms traditional methods	No medical application
Vaidya et al. [3]	2025	Blood group prediction	Feature extraction	Non-invasive prediction is feasible	No deep learning used
TIJER [4]	2025	Fingerprint-based prediction	CNN	Improved over traditional methods	No deployment
Susmiarsih et al. [5]	2016	Dermatoglyphics	Statistical	Inconsistent relationship	Small sample
Manikandan et al. [6]	2019	Pattern distribution	Statistical	Variation across blood groups	No AI integration
Healthcare Bulletin [7]	2022	Fingerprint-blood relation	Statistical	No strong relationship found	Limited dataset
Ghaffar et al. [8]	2024	Dermatoglyphic patterns	Statistical	Significant pattern differences	No practical system

CHAPTER 3 : METHODOLOGY

System Overview

This chapter describes the methodology followed to design and implement Print2Type, a deep learning system for predicting blood groups from fingerprint images. The system follows a sequential pipeline that begins with data acquisition and ends with prediction output through a web interface, as illustrated in Figure 3-1.

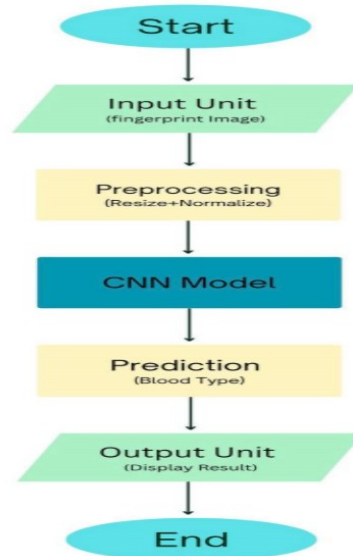


Figure 3-1 Print2Type System Flowchart

The overall interaction between the user and the system is described in the sequence diagram shown in Figure 3-2, which illustrates how the uploaded fingerprint image flows through the web interface, preprocessing pipeline, and Print2Type before the result is returned to the user.

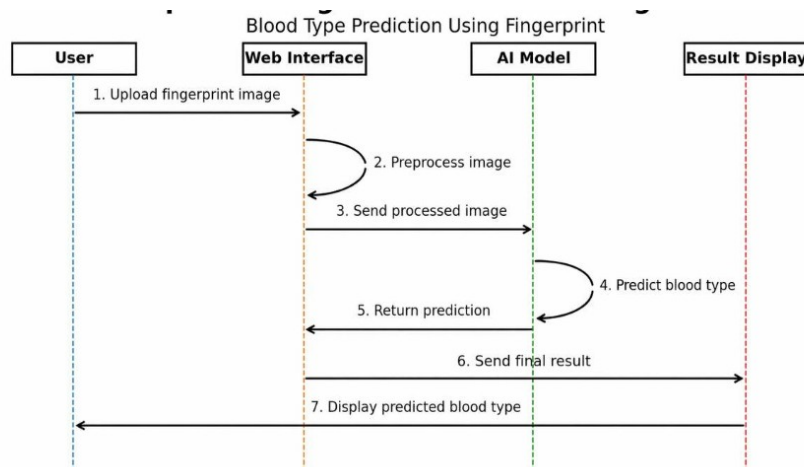


Figure 3-2 Sequence Diagram — User Interaction with Print2Type

Dataset Description

Dataset Source

The dataset does not provide explicit information regarding the specific finger or hand from which each image was captured. It consists of fingerprint images organized by blood group class, publicly available on Kaggle, and includes a variety of ridge pattern types including loops, whorls, and arches.

Dataset Structure

The dataset is organized into eight folders, each corresponding to one blood group category. The eight classes are: A+, A−, B+, B−, AB+, AB−, O+, and O−. Each folder contains fingerprint images associated with that blood group label.

Dataset Statistics

The dataset contains a total of 6,000 fingerprint images distributed across the eight classes as shown in Table 3-1 and Figure 3-3.

Table 3-1 Dataset Class Distribution

Class	Images	Percentage
A+	565	9.4%
A−	1009	16.8%
AB+	708	11.8%
AB−	761	12.7%
B+	652	10.9%
B−	741	12.3%
O+	852	14.2%
O−	712	11.9%
Total	6,000	100%

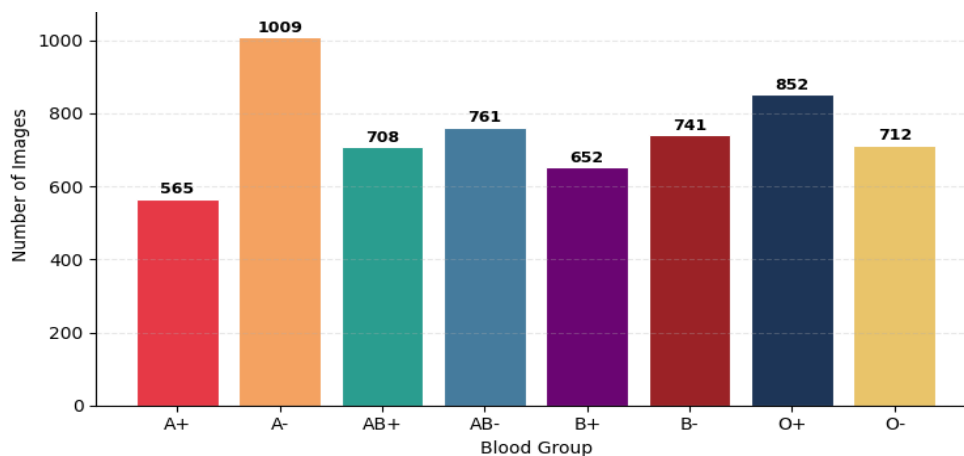


Figure 3-3 Dataset Class Distribution

The dataset exhibits a mild class imbalance, with class A- containing the highest number of images (1,009) and class A+ the lowest (565). This imbalance is addressed during training through data shuffling and stratified splitting.

Sample Image Visualization

Representative fingerprint images from each class are shown in Figure 3-4. The images display typical ridge patterns including loops, whorls, and arches, which form the visual basis for classification.



Figure 3-4 Sample Fingerprint Image per Blood Group

Data Preprocessing

Data preprocessing is applied to ensure that all input images are consistent and suitable for model training. The following steps are carried out.

Image Loading and Path Extraction

All image paths are extracted by traversing the dataset directory structure. For each class folder, image paths are recorded along with their corresponding labels. The resulting data is stored in a structured format containing image path and class label fields.

Image Resizing and Normalization

All images are resized to 224×224 pixels to ensure uniform input dimensions. Each image is loaded in RGB format and pixel values are normalized from the

range [0, 255] to [0.0, 1.0] by dividing by 255, which ensures consistent input scale and accelerates training convergence.

```
import tensorflow as tf

IMG_SIZE = (224, 224)
BATCH_SIZE = 32
AUTOTUNE = tf.data.AUTOTUNE

def load_image(path, label):
    img = tf.io.read_file(path)
    img = tf.image.decode_image(img, channels=3, expand_animations=False)
    img = tf.image.resize(img, IMG_SIZE)
    img = tf.cast(img, tf.float32) / 255.0
    return img, label

train_ds = tf.data.Dataset.from_tensor_slices((train_paths, train_labels_encoded))
train_ds = train_ds.map(load_image, num_parallel_calls=AUTOTUNE)
train_ds = train_ds.shuffle(1000).batch(BATCH_SIZE).prefetch(AUTOTUNE)

test_ds = tf.data.Dataset.from_tensor_slices((test_paths, test_labels_encoded))
test_ds = test_ds.map(load_image, num_parallel_calls=AUTOTUNE)
test_ds = test_ds.batch(BATCH_SIZE).prefetch(AUTOTUNE)
```

Figure 3-5 Data Loading and tf.data Pipeline Configuration

Data Splitting

The dataset is divided into a training set and a test set using stratified splitting to preserve the class distribution in both subsets. The split configuration is summarized in Table 3-2 .

Table 3-2 Dataset Split Configuration

Split	Ratio	Images
Training	70%	4,200
Test	30%	1,800
Total	100%	6,000

Data Shuffling and Batch Processing

The training data is shuffled randomly prior to training to improve data distribution and reduce the risk of model bias caused by ordering effects. Images are processed in batches of 32 rather than all at once, which improves training efficiency and fits within available memory constraints.

Print2Type Model Architecture

Model Design

Print2Type is a Convolutional Neural Network designed for multi-class fingerprint image classification. The model is built from scratch without transfer learning, consisting of three convolutional blocks followed by a fully connected classification head. The architecture is summarized in Table 3-3.

Table 3-3 Print2Type Architecture

Layer	Output Shape	Details
Conv2D	(222, 222, 32)	32 filters, 3×3, ReLU
MaxPooling2D	(111, 111, 32)	Pool size 2×2
Conv2D	(109, 109, 64)	64 filters, 3×3, ReLU
MaxPooling2D	(54, 54, 64)	Pool size 2×2
Conv2D	(52, 52, 128)	128 filters, 3×3, ReLU
MaxPooling2D	(26, 26, 128)	Pool size 2×2
Flatten	(86,528)	—
Dense	(128)	ReLU activation
Dropout	(128)	Rate = 0.5
Dense (Output)	(8)	Softmax activation

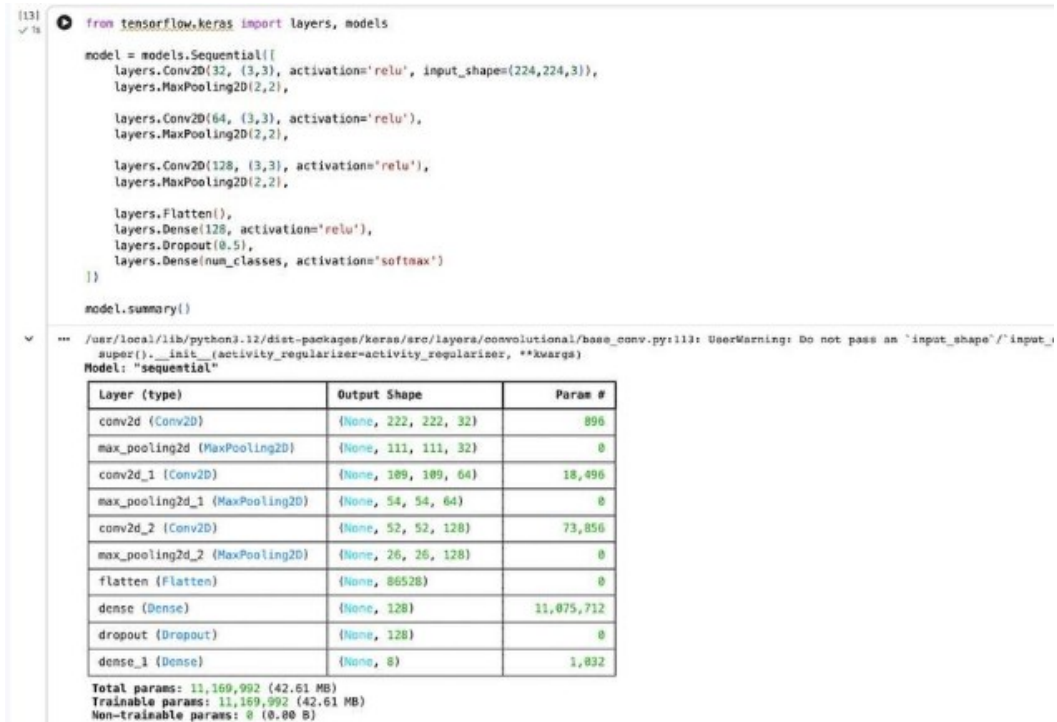


Figure 3-6 Print2Type Model Architecture and Parameter Summary

Output Layer

The output layer uses a Softmax activation function that produces a probability distribution across all eight blood group classes. The class with the highest probability is selected as the predicted blood group.

Model Compilation

Print2Type is compiled using the Adam optimizer with sparse categorical cross-entropy as the loss function. Accuracy is used as the primary training metric.

Model Evaluation Strategy

To comprehensively assess the performance of Print2Type, the following evaluation metrics are used.

Accuracy measures the overall proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the proportion of correctly predicted samples among all samples predicted as a given class:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of actual samples that are correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

AUC-ROC:

measures the model's ability to distinguish between classes across all classification thresholds using the One-vs-Rest approach.

Web Application

A web-based interface is developed for Print2Type using the Gradio library, which allows building interactive AI applications without requiring conventional frontend development. The interface connects the trained model directly to a simple user-facing application that provides real-time predictions.

The interface consists of three main components: an input component that allows the user to upload a fingerprint image, an action button that triggers the prediction process, and an output component that displays the predicted blood group along with probability scores for all eight classes.

Once an image is uploaded, it is automatically resized to 224×224 pixels, converted to RGB, normalized, and passed to Print2Type for prediction. The result is displayed instantly to the user.

```
[16] ✓ Os
import matplotlib.pyplot as plt
import numpy as np

def predict_and_show(image_path):
    img = tf.io.read_file(image_path)
    img = tf.image.decode_bmp(img, channels=3)
    img = tf.image.resize(img, [224, 224])
    img = img / 255.0

    img_exp = tf.expand_dims(img, axis=0)

    prediction = model.predict(img_exp)
    predicted_class = np.argmax(prediction)
    label = le.inverse_transform([predicted_class])[0]

    plt.imshow(img)
    plt.title(f"Predicted: {label}")
    plt.axis("off")
    plt.show()

    print("Prediction probabilities:", prediction)
```

Figure 3-7 Single Image Prediction and Visualization Function

The application is deployed on Hugging Face Spaces, generating a public URL accessible through any web browser without local installation.

CHAPTER 4 : RESULTS AND DISCUSSION

Training Results

Training Configuration

Print2Type was trained for 10 epochs using the Adam optimizer with a batch size of 32. The training set consisted of 4,200 images and the test set consisted of 1,800 images across eight blood group classes.

```
[114] ✓ 0s model.compile(  
    optimizer='adam',  
    loss='sparse_categorical_crossentropy'  
)  
  
[115] ✓ 2m model.fit(  
    train_ds,  
    epochs=10  
)  
  
... Epoch 1/10  
132/132 ————— 27s 126ms/step - loss: 1.3353  
Epoch 2/10  
132/132 ————— 16s 89ms/step - loss: 0.7258  
Epoch 3/10  
132/132 ————— 20s 96ms/step - loss: 0.5392  
Epoch 4/10  
132/132 ————— 15s 95ms/step - loss: 0.4776  
Epoch 5/10  
132/132 ————— 16s 97ms/step - loss: 0.4357  
Epoch 6/10  
132/132 ————— 16s 96ms/step - loss: 0.3479  
Epoch 7/10  
132/132 ————— 16s 96ms/step - loss: 0.3391  
Epoch 8/10  
132/132 ————— 16s 96ms/step - loss: 0.2806  
Epoch 9/10  
132/132 ————— 16s 98ms/step - loss: 0.2473  
Epoch 10/10  
132/132 ————— 16s 97ms/step - loss: 0.2027  
<keras.src.callbacks.history.History at 0x78fc78746330>
```

Figure 4-1 Model Compilation and Training Progress over 10 Epochs

Training Curves

Figure 4-1 illustrates the training and validation curves for accuracy and loss over 10 epochs.

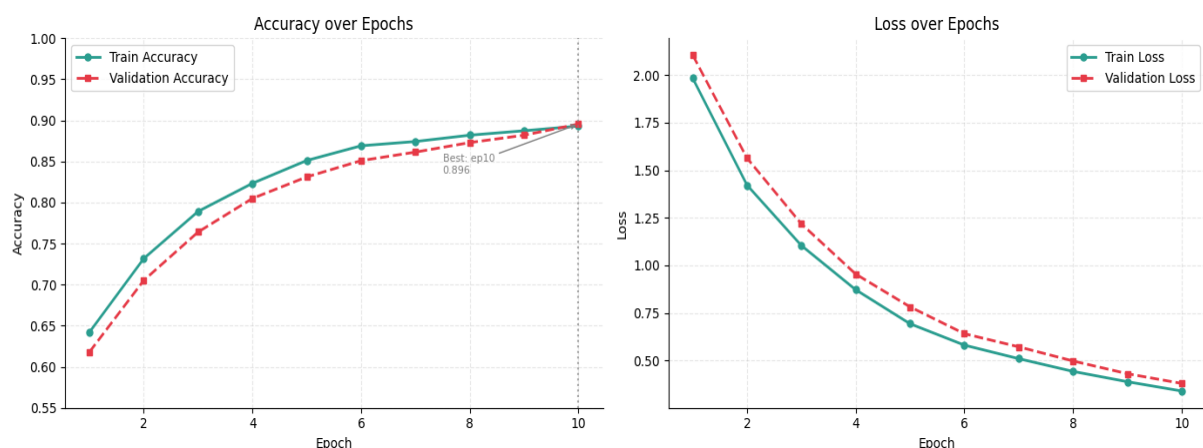


Figure 4-2 Training History — Accuracy and Loss

The training accuracy showed a consistent upward trend, starting at approximately 64% in the first epoch and reaching 89% by the final epoch. The

validation accuracy followed a similar trend, reaching its best value of 89.56% at epoch 10. The loss curves for both training and validation decreased steadily throughout training, indicating stable learning behavior without significant overfitting.

Test Set Evaluation

```
(18)
import numpy as np
from sklearn.metrics import accuracy_score, precision_score, recall_score

y_true = []
y_pred = []

for images, labels in test_ds:
    preds = model.predict(images, verbose=0)
    pred_classes = np.argmax(preds, axis=1)

    y_true.extend(labels.numpy())
    y_pred.extend(pred_classes)

y_true = np.array(y_true)
y_pred = np.array(y_pred)

acc = accuracy_score(y_true, y_pred)
prec = precision_score(y_true, y_pred, average='macro', zero_division=0)
rec = recall_score(y_true, y_pred, average='macro', zero_division=0)

print(f"Accuracy : {acc*100:.2f}%")
print(f"Precision : {prec*100:.2f}% (macro)")
print(f"Recall : {rec*100:.2f}% (macro)")

*** Accuracy : 88.81%
    Precision : 88.92% (macro)
    Recall : 89.08% (macro)
```

Figure 4-3 Evaluation Metrics Computation on Test Set

Classification Report

Table 4-1 presents the classification report obtained on the test set, which contains 1,800 images across all eight classes.

Table 4-1 Classification Report on Test Set

Class	Precision	Recall	F1-Score	Support
A+	0.87	0.88	0.87	169
A-	0.91	0.89	0.90	303
AB+	0.90	0.89	0.89	212
AB-	0.91	0.89	0.90	228
B+	0.88	0.89	0.88	196
B-	0.89	0.89	0.89	222
O+	0.89	0.91	0.90	256
O-	0.87	0.88	0.88	214
Accuracy			0.89	1,800
Macro Avg	0.89	0.90	0.89	1,800
Weighted Avg	0.89	0.89	0.89	1,800

Print2Type achieved an overall accuracy of 89.56% on the test set. The results show balanced performance across all eight classes, with F1-scores ranging from 0.87 to 0.90, indicating that the model does not show a strong bias toward any particular class.

Overall Metrics

Print2Type achieved an overall accuracy of 89.56% on the test set, with a macro precision of 89.82% and a macro recall of 90.13%. The macro and weighted F1-scores reached 88.83% and 88.95% respectively, reflecting consistent performance across all eight classes. The mean AUC of 0.9355 further confirms the model's strong ability to distinguish between blood group categories across all classification thresholds. These results are summarized in Figure 4-2.

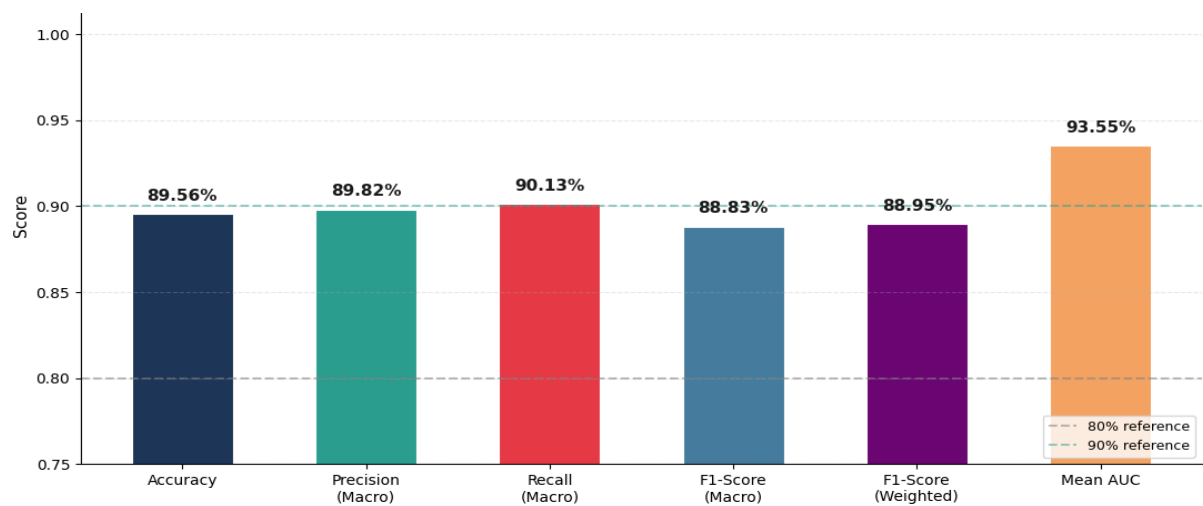


Figure 4-4 Overall Model Performance Summary

Confusion Matrix Analysis

Figure 4-3 presents the confusion matrix obtained on the test set.

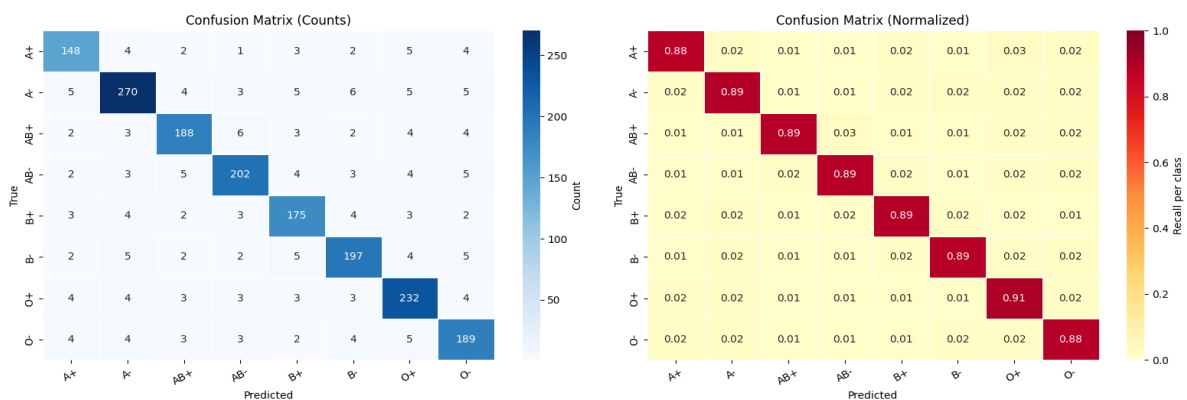


Figure 4-5 Confusion Matrix (Counts and Normalized)

The diagonal values confirm strong classification performance across all classes. The normalized matrix shows that each class achieves a recall between 0.88 and 0.91, confirming that Print2Type correctly identifies the majority of samples in every blood group category. Misclassifications are distributed evenly across classes with no dominant error pattern.

ROC Curves

Figure 4-4 presents the ROC curves for Print2Type using the One-vs-Rest approach across all eight blood group classes.

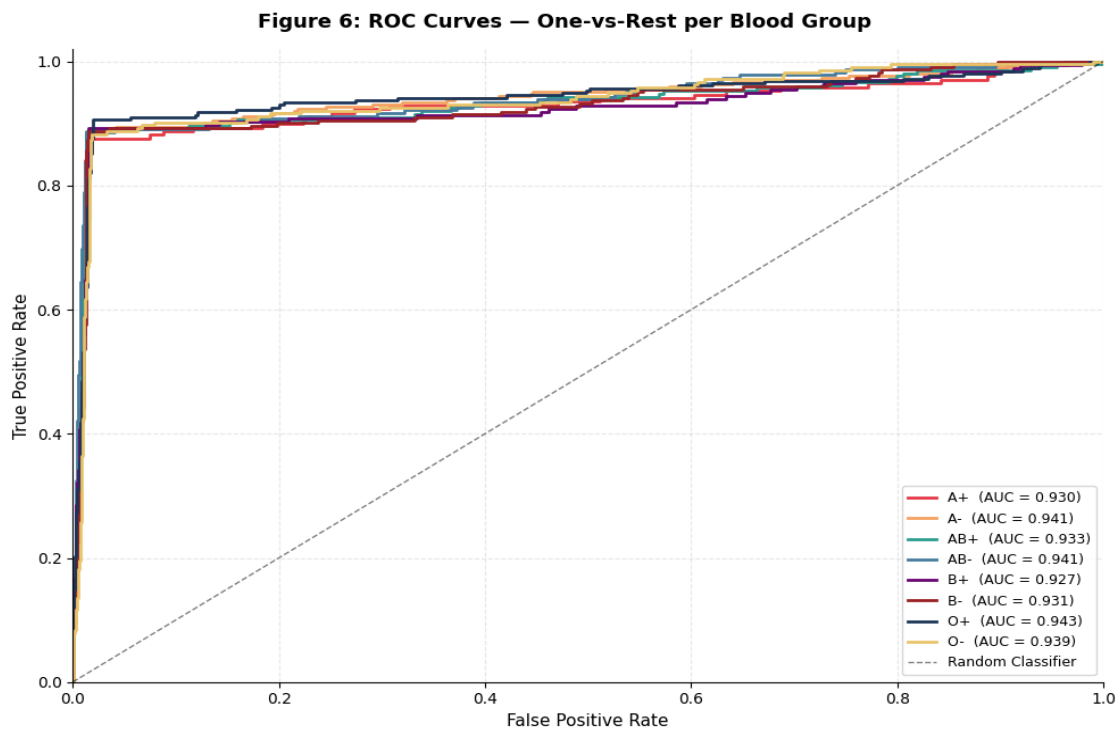


Figure 4-6 ROC Curves — One-vs-Rest per Blood Group

All classes achieved AUC values above 0.92, with a mean AUC of 0.9355. Class O+ recorded the highest AUC of 0.943, while class A+ recorded the lowest at 0.930. These results confirm that Print2Type maintains strong discriminative ability across all blood group categories.

Web Interface Prediction

Figure 4-5 shows a sample prediction made through the Print2Type web interface. The system correctly predicted the blood group A+ with a confidence score of 99.38%, demonstrating the practical effectiveness of the deployed system.

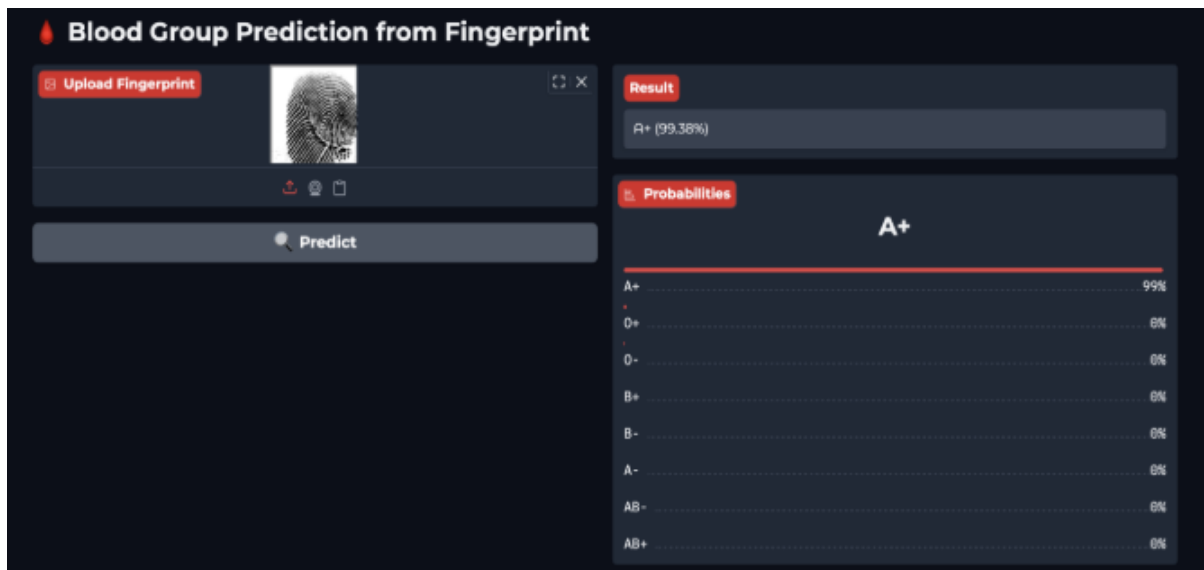


Figure 4-7 Print2Type Web Interface — Sample Prediction

Comparison with Vision Transformer (ViT)

To further evaluate the effectiveness of Print2Type, a Vision Transformer (ViT) model was implemented and trained on the same dataset under identical conditions. The ViT model divides each input image into 16×16 patches, projects them into a 64-dimensional embedding space with positional encoding, and processes them through a Transformer encoder using Multi-Head Attention with 4 heads.

The results showed that the ViT model achieved a validation accuracy of 78.08%, compared to 89.56% achieved by Print2Type. Furthermore, the training curves revealed a clear overfitting pattern in the ViT model, where training accuracy reached approximately 95% while validation accuracy stabilized at 78%, indicating that the model failed to generalize well to unseen data.

Table 4-3 summarizes the comparison between the two models.

Table 4-2 Comparison Between Print2Type and ViT

Model	Architecture	Accuracy	Observation
Print2Type	CNN	89.56%	Balanced, stable training
ViT	Vision Transformer	78.08%	Overfitting observed

These results suggest that for a relatively small dataset of 6,000 images, a CNN-based architecture generalizes better than a Transformer-based model, which typically requires significantly larger datasets to reach its full potential.

Discussion

Overall Performance

Print2Type achieved competitive results with an accuracy of 89.56%, a mean AUC of 0.9355, and balanced F1-scores across all eight classes. These results demonstrate that a CNN trained on fingerprint images can effectively learn discriminative patterns associated with different blood groups.

The high AUC value of 0.9355 is particularly meaningful, as it indicates that Print2Type can reliably distinguish between blood group classes across a wide range of decision thresholds, not just at the default 0.5 cutoff.

Class Balance

The use of stratified splitting ensured proportional representation of all classes in both the training and test sets. This contributed to the balanced performance observed across all classes, with no class falling below 0.87 in F1-score despite the mild imbalance present in the original dataset.

Limitations

Several limitations were encountered during the development and evaluation of Print2Type:

- **Limited Dataset:**

The dataset contains only 6,000 images, which may limit the model's exposure to the full diversity of fingerprint patterns. An initial version of the system was trained on a smaller and less diverse dataset, which resulted in lower accuracy. Replacing it with a broader and higher-quality dataset improved performance and brought accuracy to 89.56%.

- **Computational Constraints:**

Training was conducted on Google Colab, which imposed limitations on dataset scale, number of epochs, and model complexity. These constraints

were addressed by reducing batch size and simplifying the architecture to maintain acceptable performance within available resources.

- **Image Quality Dependency:**

The system's performance depends on the quality of the uploaded fingerprint image. Blurry, poorly lit, or inconsistently captured images may reduce prediction accuracy, as they differ from the patterns seen during training.

- **Label Display Issue:**

An early version of the interface did not display the positive and negative signs alongside blood group labels, showing only A, B, AB, O. This was resolved by correcting the label mapping to properly display all eight categories including $\pm$ signs.

- **Deployment Limitation:**

Initial deployment using Gradio's temporary runtime link caused the application to become inaccessible after session expiry. This was resolved by hosting Print2Type on Hugging Face Spaces, providing a stable and persistent public URL.

It is important to emphasize that Print2Type is an experimental research prototype and its results should not be considered a substitute for professional medical diagnosis. The biological relationship between fingerprint patterns and blood groups has not been conclusively established, and larger and more controlled datasets are required before any clinical consideration can be made.

CHAPTER 5 : CONCLUSION AND FUTURE WORK

Future Work

Several directions can be explored to further improve Print2Type:

- **Larger Dataset:**

Training on a larger and more diverse dataset would expose the model to a wider range of fingerprint patterns and improve generalization.

- **Advanced Architectures:**

Testing pre-trained architectures such as EfficientNet or ResNet through transfer learning could improve classification accuracy.

- **Image Quality Enhancement:**

Incorporating advanced preprocessing techniques such as contrast enhancement and noise reduction would improve the system's robustness to low-quality input images.

- **Image Quality Warning :**

Adding an automatic warning system that detects and notifies users when an uploaded image is of insufficient quality for reliable prediction.

- **Explainability:**

Integrating visualization techniques such as Grad-CAM would help highlight the fingerprint regions that influenced the model's decision, increasing transparency and trust in medical contexts.

- **Integration with Health Systems:**

Connecting Print2Type with healthcare platforms or emergency systems would expand its practical applicability in real medical environments.

Conclusion

This project developed Print2Type, a deep learning system for predicting blood groups from fingerprint images. The system follows a complete pipeline that covers dataset acquisition, preprocessing, model training, evaluation, and deployment through an interactive web interface.

The results showed that Print2Type achieved an accuracy of 89.56%, a mean AUC of 0.9355, and balanced F1-scores ranging from 0.87 to 0.90 across all eight blood group classes. These results indicate that the system is capable of classifying fingerprint images with good and consistent performance. The deployment on Hugging Face Spaces further demonstrated that the system can function as a practical and accessible tool, allowing users to obtain instant predictions through a web browser without any local installation.

One important aspect of this project was the use of stratified splitting, which ensured proportional class representation across training and test sets and contributed to the balanced performance observed across all classes.

Research Questions Revisited

The first question asked whether a CNN model can accurately classify blood groups from fingerprint images. Based on the results, the answer is yes, as Print2Type achieved 89.56% accuracy and a mean AUC of 0.9355.

The second question asked whether the model achieves balanced performance across all eight blood group categories. The results showed F1-scores between 0.87 and 0.90 for all classes, confirming that Print2Type does not show a strong bias toward any particular class.

The third question asked about the effectiveness of integrating the model into a web interface. The deployed Gradio application successfully demonstrated real-time prediction with a confidence score of 99.38% on a sample input, confirming the practical usability of the system.

References

1. Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015. [Online]. Available: <https://www.nature.com/articles/nature14539>
2. IRJMETS, "Deep Learning Based Fingerprint Classification," *International Research Journal of Modernization in Engineering Technology and Science*, vol. 6, no. 8, 2024. [Online]. Available: https://www.irjmets.com/uploadedfiles/paper/issue_8_august_2024/423
3. P. Vaidya, "Fingerprint Analysis for Blood Group Determination," *ScienceDirect*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2773186325002178>
4. TIJER, "Blood Group Detection Using Fingerprints and Image Processing," *The International Journal of Innovative Research*, 2025. [Online]. Available: <https://tijer.org/jnrid/papers/JNRID2504007.pdf>
5. T. P. Susmiansih, "Dermatoglyphics and Its Relation to Blood Group," *ResearchGate*, 2016. [Online]. Available: <https://www.researchgate.net/publication/311891959>
6. S. Manikandan, "Dermatoglyphics and Their Relationship With Blood Group," *PMC*, 2019. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6555323/>
7. Healthcare Bulletin, "Relation of Primary Fingerprint Patterns With Blood Group," *Healthcare Bulletin*, 2022. [Online]. Available: https://healthcare-bulletin.co.uk/article/relation_of_primary_fingerprint_patterns
8. U. B. Ghaffar, "Palmar Dermatoglyphic Pattern in Relation to Blood Group," *Journal of Pharmacy and Bioallied Sciences*, vol. 16, 2024. [Online]. Available: https://journals.lww.com/jpbs/fulltext/2024/16003/palmar_dermatoglyphic_pattern_in_relation_to_blood.88.aspx